Roll Number												

Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023
END SEMESTER EXAMINATION - ODD 2023
M(IT)101 - Engineering Mathematics - I

TIME ALLOTTED: 3Hours **FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. $\sum u_n$ and $\sum v_n$ be two series of positive real numbers , such that $lim \frac{u_n}{v_n} = 5$, and $\sum v_n$ is divergent then $\sum u_n$ is (1) CO1 BL1

a) Convergent
b) Divergent
c) Oscillatory
d) None of these

2. The system of equations $x+2y=5, 2x+4y=7$ has (1) CO4 BL4
a) unique solution
b) No solution
c) Infinite number of solutions
d) None of these

3. If A= $[a_{ij}]_{5x7}$, then, (1) CO4 BL4
a) Rank(A) >5
b) Rank(A) <5
c) Rank(A) ≤ 5
d) None of these

4. The critical point of the function f(*x,y*)=*xy* is (1) CO2 BL2
a) (1,1)
b) (-1,1)
c) (1,-1)
d) (0,0)

5. If A= $\begin{pmatrix} 5 & 2 & 3 \\ 1 & 0 & 2 \\ 0 & 4 & 0 \end{pmatrix}$, then the sum of the Eigen values of A is (1) CO2 BL2

a) 3

b) 5
c) 6
d) 10

6. If $\lambda^3 - 6\lambda^2 + 9\lambda = 4$ is the characteristic equation of a square matrix A, then A^{-1} is (1) CO2 BL2

a) $A^2 - 6A + 9I$

b) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}I$

c) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}$

d) $A^2 - 6A + 9$

7. The pair of the functions which satisfy the conditions of Cauchy's M.V.T. in [-1,1] is (1) CO2 BL2

a) $\sqrt[5]{x}, x^2$

b) $\frac{1}{\sqrt{x}}, \sqrt{x}$

c) $|x|, x$

d) $\sqrt[5]{x+2}$, x^2

8. Identify the correct statements (1) CO4 BL2

a) A bounded monotonic sequence is divergent

b) A bounded monotonic sequence is convergent

c) A bounded monotonic sequence is oscillatory

d) A bounded monotonic sequence may not have finite limit

9. Which of the following theorem can be applied on f(x)=IxI in the interval [-1, 1] (1) CO4 BL4

a)Rolle's theorem
b) Lagrange Mean value theorem
c) Cauchy Mean value theorem
d) None of these

10. The Jacobian of the two functions f(x,y)=x and g(x,y) = xy is (1) CO4 BL3

(a) y

(b) x

(c) 0

(d) none of these

11. Let A be a square matrix of order 4.The eigen values of A are 1, 2 ,3, 4.Then the determinant of A is (1) CO2 BL2

a) 24
b) 20
c) 15
d) none of these

12. If A = $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$. Then the eigen values of A is (1) CO1 BL1

a) 2,2,5
b) 4,4,5
c) 3,5,2
d) none of these

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

13. Find maxima and minima of the function f(x, y)=$x^3 + y^3 - 12x - 3y + 24$, also find the saddle points. (5) CO2 BL2

14. Evaluate the double integral $\int_0^{\pi/2} \int_{\pi/2}^{\pi} e^x \cos(y - x)\, dy\, dx$ (5) CO2 BL2

15. Verify Cayley-Hamilton theorem for the matrix $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$ and find its inverse . (5) CO2 BL1

16. Test the convergence of the series $\frac{1+2}{2^3} + \frac{1+2+3}{3^3} + \frac{1+2+3+4}{4^3} + \ldots$ (5) CO1 BL1

17. Show that the series $\sum u_n$ where $u_n = \frac{1}{4^n+3}$ is convergent. (5) CO3 BL2

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

18a. If u=$\log(x^3 + y^3 + z^3 - 3xyz)$, then show that (5) CO2 BL2

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{-3}{(x+y+z)^2}$$

18b. Examine whether the function (5) CO4 BL4

$$f(x,y) = \begin{cases} \frac{x^3 + y^3}{x^3 - y^3} & when\ (x,y) \neq (0,0) \\ 0 & when\ (x,y) = (0,0) \end{cases}$$

is continuous at $(0,0)$.

18c. If $f(x,y)=\frac{x+y}{1-xy}$ and $g(x,y)=\tan^{-1}x+\tan^{-1}y$, find the Jacobean $\frac{\partial(f,g)}{\partial(x,y)}$. (5) CO2 BL2

19a. Find the extreme values of the function $f(x,y)= x^3+y^3-63(x+y)-12xy$. (5) CO2 BL2

19b. $\iint_R xy(x+y)\,dxdy$ (5) CO4 BL4

R being the region bounded by $y^2=x$ and $y=x$

19c. If $u=u\left(\frac{y-x}{xy},\frac{z-x}{zx}\right)$ show that $x^2\frac{\partial u}{\partial x}+y^2\frac{\partial u}{\partial y}+z^2\frac{\partial u}{\partial z}=0$ (5) CO2 BL2

20a. Verify Cauchy's Mean Value Theorem for the following pair of functions (4) CO2 BL2

$f(X) = \sqrt{x}$, $g(x)= 1/\sqrt{x}$; $x \in [1, 2]$

20b. Examine whether Rolle's Theorem is applicable for $y=|x|$ in $[-1, 1]$. (5) CO4 BL4

20c. Apply Lagrange's Mean Value Theorem to prove, (6) CO3 BL3

$\frac{x}{1+x} < \log(1+x) < x$, for all $x>0$

21a. If $u=\tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, then by Euler's theorem prove (5) CO3 BL3

$(i)\; x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$

$(ii)\; x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\sin 2u(1-4\sin^2 u)$

21b. Find the eigen values and the corresponding eigen vectors of the matrix $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$ (6) CO2 BL2

21c. Evaluate (4) CO2 BL2

$$\int_0^{\frac{\pi}{2}}\int_0^{\pi}\sin(x+y)\,dxdy$$

R being the region bounded by $y^2=x$ and $y=x$.

22. (15) CO1 BL1

a) Prove that $\frac{x}{1+x^2} < tan^{-1}x < x, 0 < x < \frac{\pi}{2}$.

b)Find the extrema(maxima ,minima) of the function

$x^4+y^4-2x^2+4xy-2y^2$ and also find the saddle point if exists.

c) Verify the Lagrange's mean value theorem for the function
F(x)=x(x-1)(x-2) , $0 \leq x \leq \frac{1}{2}$ (5+8+2)
